

CHE 201 Fall 2019 HW 2

Sarah Pabilic

TOTAL POINTS

82 / 100

QUESTION 1

1 Question 1 19 / 20

✓ - 0 pts Correct

- 2 pts Did not state if kinetic energy is increasing or decreasing

- 1 pts Did not state what would be the effect of friction on DeltaKE

- 2 pts $h = -1.5$; thus $\Delta PE = -5.88 \text{ J}$ and ΔKE is 6.1 J

- 5 pts Did not take work into consideration to calculate ΔKE . Change in Kinetic Energy = 6.1 J (increasing).

- 5 pts Did not convert force into energy (Work)

- 3 pts Did not evaluate the effect of friction on

ΔKE

- 1 pts $W = 0.05 \cdot 4.39 = 0.219 \text{ J}$

- 2.5 pts Accounted twice for the gravitational force.

ΔKE is 6.1 J

- 2 pts Kinetic energy is increasing

- 10 pts Did not calculate ΔKE

✓ - 1 pts With friction, the change in kinetic energy (ΔKE) decreases - not the kinetic energy (KE).

QUESTION 2

2 Question 2 20 / 20

✓ - 0 pts Correct

- 1 pts Work is negative

- 1 pts $W = -20000 \text{ J} = -20 \text{ kJ}$

- 1 pts Did not state the value used for P_f

- 5 pts Did not show the solution for the integral

- 10 pts Wrong formula for work

- 3 pts Wrong unit. Work = -20 kJ

- 5 pts Wrong solution for the integral. Work = -20

kJ

- 3 pts Did not show how the integral results in -20

kJ

- 2 pts Did not show the conversion from $\text{bar} \cdot \text{m}^3$ to

kJ

- 1 pts Missing the unit for the final volume

- 3 pts Used wrong value for V_2 in b). $V_2 = 0.0333$ and $W = -20 \text{ kJ}$

- 1 pts The denominator in the work formula should be $1 - n$

- 5 pts Wrong calculation for $P_2 V_2$ ($= 29970 \text{ J}$) and $P_1 V_1$ ($= 10000 \text{ J}$). $W = -20 \text{ kJ}$

- 2 pts Wrong answer for final volume ($= 0.0333 \text{ m}^3$)

- 2 pts wrong answer for Work ($= -20000 \text{ J} = -20 \text{ kJ}$)

- 1 pts Integral solution should be $-P_2 V_2 + P_1 V_1$

QUESTION 3

3 Question 3 15 / 20

- 0 pts Correct

- 2 pts Stages 2 and 3 are inverted

- 4 pts Wrong or missing the volume at state 2, in ft^3

- 4 pts Wrong or missing work calculation for process 1-2

✓ - 4 pts Wrong or missing work calculation for process 2-3. Work 2-3 = 0 Btu

- 4 pts Wrong or missing work calculation for process 3-1

- 1 pts Sketch missing labels, axis units and/or process arrows

✓ - 1 pts Wrong sketch for process 1-2 ($PV = \text{constant}$ is a curved line)

- 3 pts W_{1-2} and W_{3-1} are inverted

- 2 pts Didn't show the integral calculation for Work 3-1

- 2.5 pts Unit conversion to Btu is not clear

- 2.5 pts Wrong or missing unit conversion to Btu

- 2 pts Wrong work formula for process 2-3

- **1 pts** Wrong conversion from in² to ft²
 - **2 pts** Didn't show the integral calculation for Work
- 1-2
- **1 pts** Volume at state 2 should be in ft³
 - **20 pts** Wrong answer

QUESTION 4

4 Question 4 8 / 20

- **0 pts** Correct
- **5 pts** calculation of "rate of conduction" was wrong
- **2 pts** $L_b \geq 11.22$
- ✓ - **10 pts** rate of conduction ?
- ✓ - **2 pts** thickness of the wall?
- **5 pts** thickness of the wall calculation
- **20 pts** Question 4 missing

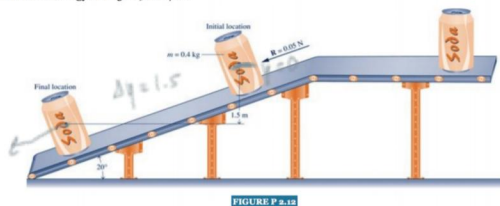
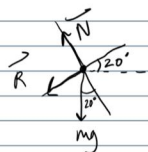
QUESTION 5

5 Question 5 20 / 20

- ✓ - **0 pts** Correct
- **20 pts** Question 5 missing

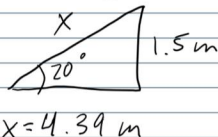
1. (20 points)

2.12 During the packaging process, a can of soda of mass 0.4 kg moves down a surface inclined 20° relative to the horizontal, as shown in Fig. P2.12. The can is acted upon by a constant force R parallel to the incline and by the force of gravity. The magnitude of the constant force R is 0.05 N. Ignoring friction between the can and the inclined surface, determine the can's change in kinetic energy, in J, and whether it is increasing or decreasing. If friction between the can and the inclined surface were significant, what effect would that have on the value of the change in kinetic energy? Let $g = 9.8 \text{ m/s}^2$.

a) ΔKE (J)?b) Friction effect on ΔKE ?

$$\Delta KE + \Delta PE + \Delta U = \cancel{Q} - W \quad \text{*assuming no heat lost/gained (part a)}$$

$$\Delta KE = \int_{s_1}^{s_2} F ds$$



$$\frac{1}{2} m (v_2^2 - v_1^2) = \int_{s_1}^{s_2} F ds \quad (\text{work}) \quad \Delta PE = 0.4 \text{ kg} (9.8 \text{ m/s}^2) (-1.5 \text{ m}) = -5.88 \text{ J}$$

$$W = 0.05 \text{ N} \cdot 4.39 \text{ m} = -0.2195 \text{ J}$$

$$\Delta KE + \Delta PE = -W$$

$$\Delta KE = -W - \Delta PE$$

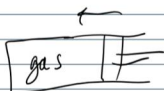
$$\Delta KE = 6.10 \text{ J, so it is increasing}$$

If friction were significant, it would oppose the force R which would affect how much of R being implemented in the motion of the can which would decrease KE

2. (20 points)

2.17 A gas in a piston-cylinder assembly undergoes a process for which the relationship between pressure and volume is $pV^2 = \text{constant}$. The initial pressure is 1 bar, the initial volume is 0.1 m^3 , and the final pressure is 9 bar. Determine (a) the final volume, in m^3 , and (b) the work for the process, in kJ.

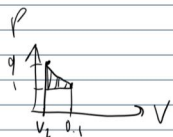
$$P_1 V_1^2 = P_2 V_2^2 \quad \leftarrow ?$$



$$P_1 V_1^2 = P_2 V_2^2 \quad \leftarrow n$$

$$\frac{1 \text{ bar} (0.1 \text{ m}^3)^2}{9 \text{ bar}} = 9 \text{ bar } V_2^2$$

$$V_2 = 0.033 \text{ m}^3$$



$$W = \frac{P_2 V_2 - P_1 V_1}{1 - 2}$$

$$= \frac{9 \text{ bar} (0.033 \text{ m}^3) - 1 \text{ bar} (0.1 \text{ m}^3)}{1 - 2} \left(\frac{10^5 \text{ N/m}^2}{1 \text{ bar}} \right) \left(\frac{1 \text{ kJ}}{10^3 \text{ N} \cdot \text{m}} \right)$$

$$= -19.7 \text{ kJ}$$

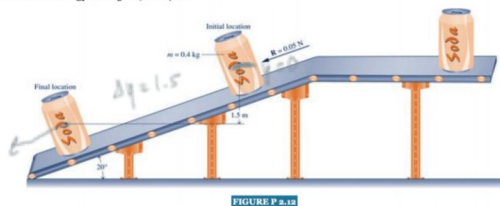
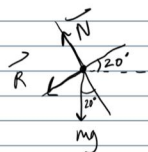
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 - 1 pts $W = 0.05 * 4.39 = 0.219 \text{ J}$
 - 2.5 pts Accounted twice for the gravitational force. ΔKE is 6.1 J
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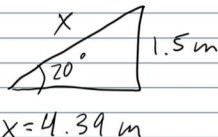
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$$W = 0.05 \text{ N} \cdot 4.39 \text{ m} = -0.2195 \text{ J}$$

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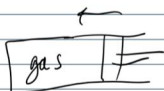
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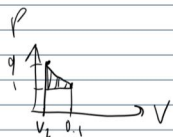
$$P_1 V_1^2 = P_2 V_2^2 \quad \leftarrow ?$$



$$P_1 V_1^2 = P_2 V_2^2 \quad \leftarrow n$$

$$\frac{1 \text{ bar} (0.1 \text{ m}^3)^2}{9 \text{ bar}} = 9 \text{ bar} V_2^2$$

$$V_2 = 0.033 \text{ m}^3$$



$$W = \frac{P_2 V_2 - P_1 V_1}{1 - 2}$$

$$= \frac{9 \text{ bar} (0.033 \text{ m}^3) - 1 \text{ bar} (0.1 \text{ m}^3)}{1 - 2} \left(\frac{10^5 \text{ N/m}^2}{1 \text{ bar}} \right) \left(\frac{1 \text{ kJ}}{10^3 \text{ N}\cdot\text{m}} \right)$$

$$= -19.7 \text{ kJ}$$

2 Question 2 20 / 20

✓ - 0 pts Correct

- 1 pts Work is negative
- 1 pts $W = -20000 \text{ J} = -20 \text{ kJ}$
- 1 pts Did not state the value used for P_f
- 5 pts Did not show the solution for the integral
- 10 pts Wrong formula for work
- 3 pts Wrong unit. Work = -20 kJ
- 5 pts Wrong solution for the integral. Work = -20 kJ
- 3 pts Did not show how the integral results in -20 kJ
- 2 pts Did not show the conversion from $\text{bar}\cdot\text{m}^3$ to kJ
- 1 pts Missing the unit for the final volume
- 3 pts Used wrong value for V_2 in b). $V_2 = 0.0333$ and $W = -20 \text{ kJ}$
- 1 pts The denominator in the work formula should be $1 - n$
- 5 pts Wrong calculation for P_2V_2 (= 29970 J) and P_1V_1 (= 10000 J). $W = -20 \text{ kJ}$
- 2 pts Wrong answer for final volume (= 0.0333 m^3)
- 2 pts wrong answer for Work (= -20000 J = -20 kJ)
- 1 pts Integral solution should be $-P_2V_2 + P_1V_1$

3. (20 points)

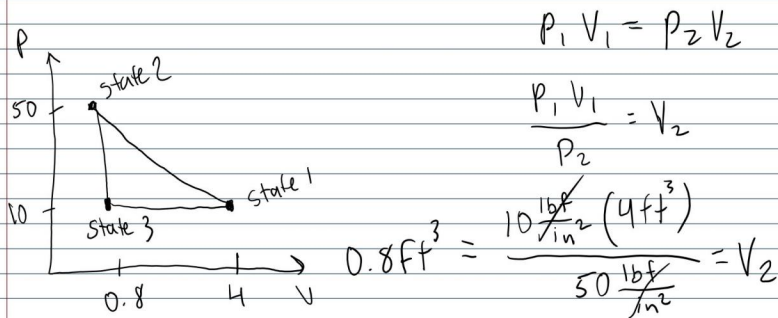
2.25 Air contained within a piston-cylinder assembly undergoes three processes in series:

Process 1-2: Compression during which the pressure-volume relationship is $pV = \text{constant}$ from $p_1 = 10 \text{ lbf/in}^2$, $V_1 = 4 \text{ ft}^3$ to $p_2 = 50 \text{ lbf/in}^2$ $V_2 = ?$

Process 2-3: Constant volume from state 2 to state 3 where $p = 10 \text{ lbf/in}^2$

Process 3-1: Constant pressure expansion to the initial state.

Sketch the processes in series on p - V coordinates. Evaluate (a) the volume at state 2, in ft^3 , and (b) the work for each process, in Btu.



$$10 \frac{\text{lbf}}{\text{in}^2} \cdot 4 \text{ ft}^3 = 40 \frac{\text{lbf} \cdot \text{ft}^3}{\text{in}^2}$$

$$W_1 = \int_4^{0.8} \frac{40}{V} dV = 40 \ln|0.8| - 40 \ln|4| = -64.4 \frac{\text{lbf} \cdot \text{ft}^3}{\text{in}^2} \left| \frac{144 \frac{\text{lbf}}{\text{ft}^2}}{1 \frac{\text{lbf}}{\text{in}^2}} \right| = -9270.4 \text{ lbf} \cdot \text{ft} \left| \frac{1 \text{ Btu}}{778.17 \text{ ft} \cdot \text{lbf}} \right|$$

$$W_1 = -11.91 \text{ Btu}$$

$$W_3 = \int_{0.8}^4 10 dV = 10 V \Big|_{0.8}^4 = 32 \frac{\text{lbf} \cdot \text{ft}^3}{\text{in}^2} \left| \frac{144 \frac{\text{lbf}}{\text{ft}^2}}{1 \frac{\text{lbf}}{\text{in}^2}} \right| = 4608 \text{ lbf} \cdot \text{ft} \left| \frac{1 \text{ Btu}}{778.17 \text{ ft} \cdot \text{lbf}} \right|$$

$$W_3 = 5.92 \text{ Btu}$$

4. (20 points)

2.36 As shown in Fig. P2.36, an oven wall consists of a 0.635-cm-thick layer of steel ($k_s = 15.1 \text{ W/m} \cdot \text{K}$) and a layer of brick ($k_b = 0.72 \text{ W/m} \cdot \text{K}$). At steady state, a temperature decrease of 0.7°C occurs over the steel layer. The inner temperature of the steel layer is 300°C . If the temperature of the outer surface of the brick must be no greater than 40°C , determine the thickness of brick, in cm, that ensures this limit is met. What is the rate of conduction, in kW per m^2 of wall surface area?

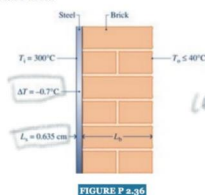


FIGURE P2.36

can solve for L_{brick}

Fourier's law:

$$Q = -kA \frac{dT}{dx}$$

↑
area

↑
thickness of material

Assume = wall is steady state
temp. is linear

$$Q_{\text{steel}} = Q_{\text{brick}} \quad \frac{Q}{A} = \text{rate of conduction}$$

$$-15.1 \text{ W/m} \cdot \text{K} \left(\frac{-0.7^\circ\text{C}}{0.635 \text{ cm}} \right) = -0.72 \text{ W/m} \cdot \text{K} \left(\frac{40 - 300^\circ\text{C}}{L_b} \right)$$

3 Question 3 15 / 20

- 0 pts Correct
- 2 pts Stages 2 and 3 are inverted
- 4 pts Wrong or missing the volume at state 2, in ft³
- 4 pts Wrong or missing work calculation for process 1-2
- ✓ - 4 pts Wrong or missing work calculation for process 2-3. Work 2-3 = 0 Btu
- 4 pts Wrong or missing work calculation for process 3-1
- 1 pts Sketch missing labels, axis units and/or process arrows
- ✓ - 1 pts Wrong sketch for process 1-2 (PV = constant is a curved line)
- 3 pts W1-2 and W3-1 are inverted
- 2 pts Didn't show the integral calculation for Work 3-1
- 2.5 pts Unit conversion to Btu is not clear
- 2.5 pts Wrong or missing unit conversion to Btu
- 2 pts Wrong work formula for process 2-3
- 1 pts Wrong conversion from in² to ft²
- 2 pts Didn't show the integral calculation for Work 1-2
- 1 pts Volume at state 2 should be in ft³
- 20 pts Wrong answer

3. (20 points)

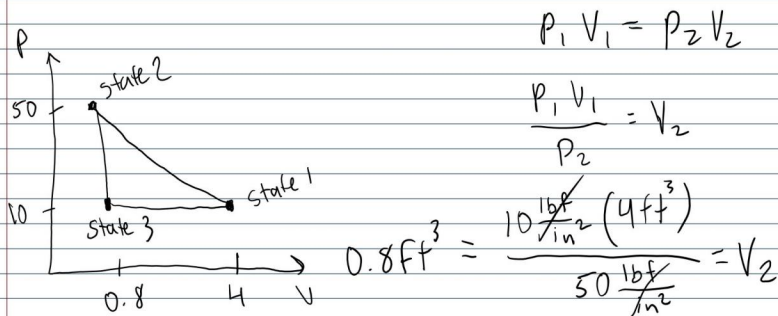
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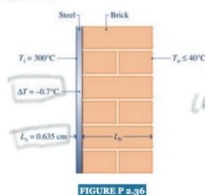


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$$L_B = \left(\frac{0.72}{15.1} \right) (40 - 300^\circ\text{C}) \left(\frac{0.635 \text{ cm}}{-0.7^\circ\text{C}} \right)$$

$$L_B = 11.25 \text{ cm}$$

$$\dot{q} = k_s \times \frac{\Delta T_s}{L_s}$$

$$= -15.1 \frac{\text{W}}{\text{m} \cdot \text{K}} \left(\frac{-0.7^\circ\text{C}}{0.635 \times 10^{-2} \text{ m}} \right) \left(\frac{1 \text{ kW}}{1000 \text{ W}} \right)$$

$$= 1.66 \frac{\text{kW}}{\text{m}^2}$$

4 Question 4 8 / 20

- 0 pts Correct
- 5 pts calculation of "rate of conduction" was wrong
- 2 pts $L_b \geq 11.22$
- ✓ - 10 pts rate of conduction ?
- ✓ - 2 pts thickness of the wall?
 - 5 pts thickness of the wall calculation
 - 20 pts Question 4 missing

Federal regulations note that when managing batteries, the container they are stored in should be “closed, structurally sound, and compatible with the batteries” (EPA.gov). The batteries or their containers must be labelled indicating that it is a waste battery. Handlers of both small quantity and large quantity universal waste may sort, mix, discharge, disassembled into individual batteries, or remove the electrolyte from the batteries and must also determine if any of the waste produced while managing the batteries is deemed hazardous (Electronic Code of Federal Regulations). While researching for state regulations for Illinois, I find that there are not many radical laws that would significantly differ from the federal ones. According to the Solid Waste Agency of Northern Cook County (SWANCC), in Illinois, singular use, non-rechargeable batteries are deemed non-hazardous and may be disposed of with normal household waste, unlike lead acid batteries which are prohibited from being disposed of in a landfill or incinerator. People or dealers that sell lead acid batteries are also responsible for the recycling of used batteries in exchange. Additionally, starting in the year 2020, “no person shall knowingly mix a lead-acid battery with any other material intended for collection as a recyclable material by a hauler” (Illinois Compiled Statutes). While researching, I was so focused on alkaline batteries and lead acid batteries that I almost completely forgot to search for the disposal regulations for rechargeable batteries. After remembering, I found that they can be recycled at most retail stores (SWANCC). Out of all my sources, I found that SWANCC had most information on how to properly dispose of batteries in one spot rather than having to search it up separately which in a sense is more effective, but that is if someone actively searched up how to dispose of batteries in

the first place. Without the prompt to search up these regulations, I would probably be oblivious to the proper procedures, given that I barely notice any other source in everyday life that could provide such information. Perhaps more could be done to help make citizens aware of these regulations through newsletters and mass emails and the like, even if it is once or twice a month so that everyone has it for common knowledge.

Resources

https://www.ecfr.gov/cgi-bin/text-idx?SID=b79008b14fb36fd528cc84f87ed9af4e&mc=true&node=pt40.27.273&rgn=div5#se40.29.273_110

<https://www.swancc.org/batteries>

<https://www.epa.gov/hw/frequent-questions-about-universal-waste>

<http://www.ilga.gov/legislation/ilcs/fulltext.asp?DocName=041500050K22.23>

5 Question 5 20 / 20

✓ - 0 pts Correct

- 20 pts Question 5 missing